

Binomial coefficients $\binom{n}{2}$ that are products of consecutive primes

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DRAFT — September 29, 2026

Abstract

Erdős and Graham asked whether $\binom{n}{k}$ can be a product of consecutive primes infinitely often (Erdős Problem #386). For $k = 2$ the known solutions are $n = 4, 6, 15, 21, 715$. We prove an elementary *balance inequality*: if $\binom{n}{2} = p_a p_{a+1} \cdots p_{a+k-1}$ with $p_a \geq 5$, then $(p_{a+k-1}/p_a)^{\lfloor k/2 \rfloor} \geq 2 - 1/p_a$. Combined with explicit short-interval results for primes, this reduces the problem to a finite computation for every bound on n . We show there is no further solution with $n \leq 10^{682}$, and with $n \leq 10^{9614}$ assuming a result of Dusart (2018). Unconditionally in n , any further solution must be a product of at least 137 consecutive primes (at least 1924 under the same result). An exhaustive search shows that no block starting at a prime below 100 and ending at a prime below $4 \cdot 10^9$ gives a new solution. The core arithmetic inequality is formalized in Lean 4.

1 Introduction

Let $p_0 = 2, p_1 = 3, \dots$ denote the primes. A *block* is a product $B(a, k) = p_a p_{a+1} \cdots p_{a+k-1}$ of $k \geq 1$ consecutive primes. Erdős and Graham [4] asked whether a binomial coefficient $\binom{n}{k}$, $2 \leq k \leq n-2$, can equal a block infinitely often. For $k = 2$ they remark that a proof that this happens only finitely often “seems hopeless”. The problem appears as Problem #386 at [5], and the case $k = 2$ is formalized as `erdos_386.variants.two` in [6].

For $k = 2$ the question is: for which $n \geq 4$ is $n(n-1)/2$ a block? The known solutions are

$$\binom{4}{2} = 2 \cdot 3, \quad \binom{6}{2} = 3 \cdot 5, \quad \binom{15}{2} = 3 \cdot 5 \cdot 7, \quad \binom{21}{2} = 2 \cdot 3 \cdot 5 \cdot 7, \quad \binom{715}{2} = 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13 \cdot 17.$$

The OEIS entry A280992 [8] reports no further solution with $n \leq 5 \cdot 10^6$. Blocks starting at 3 correspond to $n(n-1) = r\#$ (a primorial). This is the question of Nelson, Penney and Pomerance [7], who searched the first 3049 primes.

Results. Our main results are:

1. (Theorem 7) There is no solution other than the five above with $n \leq 10^{12}$ (unconditionally), with $n \leq 10^{682}$ (using Dusart 2010), and with $n \leq 10^{9614}$ (using Dusart 2018).
2. (Theorem 8) For all n : any further solution is a block of $k \geq 137$ primes (Dusart 2010), and of $k \geq 1924$ primes (Dusart 2018).
3. (Theorem 9) No block $B(a, k)$ with $p_a < 100$ and $p_{a+k-1} \leq 4 \cdot 10^9$ gives a new solution.

All three rest on the balance inequality (Lemma 2), whose proof is elementary.

2 The balance inequality

Lemma 1. For $n \geq 4$, $\binom{n}{2}$ is not prime.

Proof. $\binom{n}{2}$ equals $(n/2)(n-1)$ or $n \cdot \frac{n-1}{2}$, and both factors are at least 2. $\square$

Lemma 2 (Balance inequality). Let $n \geq 4$ and suppose $\binom{n}{2} = B(a, k)$ with $p = p_a \geq 5$. Let $r = p_{a+k-1}$ and $h = \lfloor k/2 \rfloor$. Then

$$pr^h \geq (2p-1)p^h, \quad \text{equivalently} \quad \left(\frac{r}{p}\right)^h \geq 2 - \frac{1}{p}.$$

Proof. Let E be the even and O the odd element of $\{n, n-1\}$, and put $A = E/2$. Then $\binom{n}{2} = AO$ with $\gcd(A, O) = 1$ and $|2A - O| = 1$. Every prime in the block is ≥ 5 , so AO is odd and squarefree. The block's primes therefore split as $S \sqcup T$ with $A = \prod S$ and $O = \prod T$. Since $n \geq 4$ we have $A \geq 2$ and $O \geq 3$, so S and T are non-empty. Put $s = |S|$ and $t = |T|$, so $s + t = k$, and note that every element of $S \cup T$ lies in $[p, r]$.

Case $s = t$. Then $O/A = 2 \pm 1/A \geq 2 - 1/A \geq 2 - 1/p$, while $O/A \leq r^s/p^s$.

Case $s < t$. Then $p^{s+1} \leq p^t \leq O \leq 2A + 1 \leq 2r^s + 1$, so $(r/p)^s \geq p/2 - 1/(2p^s) \geq p/2 - 1/(2p) \geq 2 - 1/p$, since $p^2 - 4p + 1 \geq 0$.

Case $s > t$. Then $2p^{t+1} \leq 2A \leq O + 1 \leq r^t + 1$, so $(r/p)^t \geq 2p - p^{-t} \geq 2 - 1/p$.

In every case $(r/p)^{\min(s,t)} \geq 2 - 1/p$. Since $\min(s, t) \leq h$ and $r \geq p$, the claim follows. $\square$

Remark 3. The inequality is sharp: $\binom{14}{2} = 91 = 7 \cdot 13$ has $p = 7$, $r = 13$, $h = 1$, and $13/7 = 2 - 1/7$. (This is not a solution, because 7 and 13 are not consecutive primes.) The proof does not use consecutiveness. We checked the general form numerically for all $n \leq 3 \cdot 10^6$.

For a fixed start a , the left-hand side of Lemma 2 is non-decreasing in k . Hence:

Corollary 4 (Monotonicity). If the inequality of Lemma 2 fails for the block $B(a, k_0)$, it fails for every $B(a, k)$ with $k \leq k_0$.

Lemma 5 (Integer test). Let $p \geq 41$, $d = r - p \geq 0$ and $h \geq 0$. If $100hd < 68p$, then $(r/p)^h < 2 - 1/p$.

Proof. $(1 + d/p)^h \leq e^{hd/p} < e^{0.68} < 1.974 < 2 - 1/41 \leq 2 - 1/p$. $\square$

3 Large starting primes

We use explicit short-interval theorems of the following form:

(G) For all $x \geq x_0$ there is a prime in $(x, x(1 + \varepsilon(x))]$, where ε is non-increasing.

Dusart [2, Prop. 6.8] proves (G) with $x_0 = 396738$ and $\varepsilon(x) = 1/(25 \ln^2 x)$. Dusart [3, Cor. 5.5] (see also [1]) proves it with $x_0 = 468991632$ and $\varepsilon(x) = 1/(5000 \ln^2 x)$. Let $X_0 = 10\,726\,905\,041$.

Lemma 6. Assume (G) with $x_0 \leq X_0$, and put $\varepsilon = \varepsilon(X_0)$. If $\binom{n}{2} = B(a, k)$ with $p_a \geq X_0$, then

$$\varepsilon(k-1)\lfloor k/2 \rfloor \geq \ln(2 - 1/X_0).$$

Proof. For $i \geq a$ we have $p_{i+1} \leq p_i(1 + \varepsilon(p_i)) \leq p_i(1 + \varepsilon)$, so $r \leq p(1 + \varepsilon)^{k-1}$ and $(r/p)^h \leq \exp(\varepsilon(k-1)h)$. Now apply Lemma 2, using $2 - 1/p \geq 2 - 1/X_0$. $\square$

Evaluating Lemma 6 in high-precision arithmetic, a solution with $p_a \geq X_0$ needs $k \geq 137$ under [2], and $k \geq 1924$ under [3].

4 Computation

Square test. $B = n(n-1)/2$ if and only if $1 + 8B = (2n-1)^2$. When B is small we test this with an exact integer square root. For huge blocks we use 40–48 filter primes ℓ and reject B as soon as $1 + 8B \bmod \ell$ is a quadratic non-residue (computed with the Jacobi-symbol algorithm). A true square is never rejected. Every block that passes all the filters is re-checked exactly.

Search A (bounded n). Fix N , $M = N(N-1)/2$ and L = the bit length of M . For a start p , let $e = \lfloor \log_2 p \rfloor$ and $K = \lceil L/e \rceil$. Blocks of length $\geq K$ exceed M . For each start $p < X_0$ with $p \geq 41$, if Lemma 5 shows that Lemma 2 fails at length $K-1$, the start is discarded (Corollary 4). Otherwise all lengths $2 \leq k < K$ are tested. Starts $p < 41$ are always tested in full. Starts $p \geq X_0$ are handled by Lemma 6 (the required k exceeds $\ln M / \ln X_0$).

Search B (bounded k). This is the same search with $K-1$ replaced by a fixed maximal length $k_{\max}$.

Validation.

- Three independent programs (a block search, a loop over n with factorization, and Search A with and without the pruning of Lemma 5) agree for $N = 10^6$ and 10^9 .
- In every run the number of starts processed equals $\pi(X_0) = 486\,570\,088$, computed independently.
- No block has ever passed all the filters without being a genuine solution.

Theorem 7. *If $4 \leq n \leq N$ and $\binom{n}{2}$ is a block, then $n \in \{4, 6, 15, 21, 715\}$, where $N = 10^{12}$ unconditionally, $N = 10^{682}$ assuming [2], and $N = 10^{9614}$ assuming [3].*

Theorem 8. *If $\binom{n}{2}$ is a block of k primes and $n \notin \{4, 6, 15, 21, 715\}$, then $k \geq 137$ (assuming [2]) and $k \geq 1924$ (assuming [3]).*

Proof. Starts $p_a \geq X_0$ are handled by Lemma 6. Search B with $k_{\max} = 136$ and $k_{\max} = 1923$ covers all starts $p_a < X_0$. $\square$

Theorem 9. *If $\binom{n}{2} = B(a, k)$ with $p_a < 100$ and $p_{a+k-1} \leq 4 \cdot 10^9$, then $n \in \{4, 6, 15, 21, 715\}$. In particular, $n(n-1) = r\#$ and $n(n-1) = 2 \cdot r\#$ have no other solutions with $r \leq 4 \cdot 10^9$.*

run	time (8 cores)	starts enumerated	blocks tested
Theorem 7, $N = 10^{12}$ (no pruning)	160 s	—	$2.7 \cdot 10^{10}$
Theorem 7, $N = 10^{682}$	33 s	45 555	$5.1 \cdot 10^6$
Theorem 7, $N = 10^{9614}$	480 s	4 702 043	$4.8 \cdot 10^9$
Theorem 8, $k_{\max} = 1923$	138 s	2 896 301	$1.9 \cdot 10^9$
Theorem 9	775 s	25	$4.7 \cdot 10^9$

Table 1: Computations (Apple M2).

5 Heuristics and obstacles

The integers n with $2B \mid n(n-1)$ are the 2^k residues obtained from the splittings of the block's primes between n and $n-1$. A solution is exactly the case where the least nontrivial residue n satisfies $\lambda := n(n-1)/(2B) = 1$. Modelling the residues as 2^k random points in $[0, 2B)$ predicts $\lambda \approx B/4^k$. For blocks starting at 2 or 3 and ending at $r \leq 97$ (exhaustive enumeration), the observed minimum λ follows this prediction up to $\lambda \approx 10^{22}$. It equals 1 only for the known solutions.

Several natural routes to finiteness seem blocked:

- **abc.** The abc conjecture gives nothing, because $n(n-1)$ is squarefree up to the factor 2. (Compare Brocard's problem, where abc does apply.)
- **Size arguments.** The equation $|2 \prod S - \prod T| = 1$ sits exactly at the trivial bound for a difference of distinct integers.
- **Equidistribution.** The Fourier transform of the residue set is the Riesz product $\prod_q (1 + e(hc_q/q))$, with $c_q \equiv (2B/q)^{-1} \pmod{q}$. The set has only 2^k points, far too few for equidistribution at scale $\sqrt{B}$ to rule out a point in $[2, \sqrt{2B}]$.

6 Formalization and data

Lemma 1 and the arithmetic core of Lemma 2 (the three cases above, stated for lists of factors in $[p, r]$), together with the monotonicity in h , are formalized in Lean 4 without Mathlib. The proofs use only the axioms `propext` and `Quot.sound`. The code, logs and Lean sources are available at [repository URL].

Acknowledgements. The computations, proofs and formalization were developed with substantial assistance from Claude (Anthropic), an AI model. The author checked [TO BE COMPLETED BY AUTHOR: describe what you personally verified].

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